Name: Toppliz ID: _____ No: _____

2110201 Quiz 2: Concept

$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 3 & 0 & 0 \end{array}\right]$

1. (5 points) True/False question. Please fill either T, F, or fill nothing for the following statements *(+1 point for correct answer, -0.5 points for wrong answer, 0 point if nothing is filled)*

   a. T A line in 3D can be written as a system of 5 different linear equations of 3 unknowns. nullity = 1 variables

   b. F Any two distinct points in 2D can be used to calculate a linear equation in the form of $y = ax + b$ where a and b are real numbers. $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix} \to \boxed{x=1}$ slope $= \infty$ a is a slope

   c. F If we pick 3 distinct points in 3D, those 3 points can ALWAYS be used to calculate a UNIQUE plane that contains those 3 points. No, when 3 points lie on the same line.

   d. F A single linear equation of 1 variable always describes a point in 1D. $\boxed{0x = c}$ $c=0$; sol $= \mathbb{R}$, $c \neq 0$; sol $= \emptyset$

   e. F Consider a plane $U = \{u_0 + a_1u_1 + a_2u_2\}$ and a plane $V = \{v_0 + a_3v_1 + a_4v_2\}$, for any vector u_i and v_i in $\mathbb{R}^4$ and any real number a_i, U and V always intersect. If $\vec{u}_1 = \vec{u}_2$, $\vec{v}_1 = \vec{v}_2$, Can select $\vec{u}_0, \vec{v}_0$ s.t. U, V are not the same plane $\therefore$ then U, V are parallel. NOT intersect!

2. (2 points) Any circle in 2D is a set of $\begin{bmatrix}x\\y\end{bmatrix}$ that satisfies this equation: $(a+x)^2 + (b+y)^2 + c = 0$ where a, b and c are real numbers. We want to find the equation for a circle that pass through these 3 points: $\begin{bmatrix}1\\1\end{bmatrix}$, $\begin{bmatrix}2\\4\end{bmatrix}$, and $\begin{bmatrix}5\\3\end{bmatrix}$. Write an augmented matrix of a system of linear equations that solve for a, b and c of this circle.

$(a+1)^2 + (b+1)^2 + c = 0$
$(a+2)^2 + (b+4)^2 + c = 0$
$(a+5)^2 + (b+3)^2 + c = 0$

$a^2 + 2a + 1 + b^2 + 2b + 1 + c = 0$
$a^2 + 4a + 4 + b^2 + 8b + 16 + c = 0$
$a^2 + 10a + 25 + b^2 + 6b + 9 + c = 0$

$$\begin{array}{c} \begin{array}{ccccc} c & a & b & a^2 & b^2 \end{array} \\ \left[\begin{array}{ccccc|c} 1 & 2 & 2 & 1 & 1 & -2 \\ 1 & 4 & 8 & 1 & 1 & -20 \\ 1 & 10 & 6 & 1 & 1 & -34 \end{array}\right] \end{array}$$

NO NEED TO DO THIS PART

$$\begin{array}{c} \begin{array}{ccccc} c & a & b & a^2 & b^2 \end{array} \\ \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 1 & 1 & 8 \\ 0 & 1 & 0 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & 0 & -2 \end{array}\right] \end{array}$$

$\therefore a = -3$, $b = -2$, $c = -5$

$c = 8 - a^2 - b^2 = 8 - 9 - 4 = -5$

3. (2 points) Find the determinant of the matrix $B = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 6 & 1 & 3 & 4 \\ 4 & 5 & 7 & 5 \\ 8 & 3 & 6 & 4 \end{bmatrix}$

$$\begin{vmatrix} 2 & 3 & 4 & 5 \\ 6 & 1 & 3 & 4 \\ 4 & 5 & 7 & 5 \\ 8 & 3 & 6 & 4 \end{vmatrix} = \begin{vmatrix} 10 & 6 & 10 & 9 \\ 10 & 6 & 10 & 9 \\ 4 & 5 & 7 & 5 \\ 8 & 3 & 6 & 4 \end{vmatrix} ; \begin{array}{l} P_4 + P_1 \\ P_3 + P_2 \end{array}$$

$= 0$ Since it has two identical rows.

Name: _____ ID: _____ No: _____

4. (4 points) We have a line and a plane in 3D. The line is the set $\left\{\begin{bmatrix}0\\2\\0\end{bmatrix} + a\begin{bmatrix}1\\3\\1\end{bmatrix} \middle| a\in\mathbb{R}\right\}$ while the plane contains these 3 points: $\begin{bmatrix}2\\0\\0\end{bmatrix}$, $\begin{bmatrix}0\\4\\0\end{bmatrix}$, and $\begin{bmatrix}0\\0\\-1\end{bmatrix}$. Find the intersection between the line and the plane.

Sol #1

Plane Equation: $ax+by+cz=d$, Substitute $\begin{bmatrix}x\\y\\z\end{bmatrix}$ with those 3 points!

$$\begin{array}{c} \begin{matrix} a & b & c & d & \end{matrix} \\ \left[\begin{array}{cccc|c} 2 & 0 & 0 & -1 & 0 \\ 0 & 4 & 0 & -1 & 0 \\ 0 & 0 & -1 & -1 & 0 \end{array}\right] \end{array} \to \left[\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 1 & 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array}\right] \quad \therefore \begin{bmatrix}a\\b\\c\\d\end{bmatrix} \in \left\{ \begin{bmatrix}\frac{1}{2}\\\frac{1}{4}\\-1\\1\end{bmatrix} d \;\middle|\; d\in\mathbb{R} \right\}$$

Let $d=4$; $\boxed{2x+y-4z=4}$

Line: $\begin{bmatrix}x\\y\\z\end{bmatrix} \in$ Line, then $\begin{cases} x=a \\ y=2+3a \\ z=a \end{cases}$

Substitute: $2(a)+(2+3a)-4(a)=4$

$\boxed{a=2}$ #

$\therefore$ Intersection $= \left\{ \begin{bmatrix}0\\2\\0\end{bmatrix} + 2\begin{bmatrix}1\\3\\1\end{bmatrix} \right\}$

$= \left\{ \begin{bmatrix}2\\8\\2\end{bmatrix} \right\}$ #

Sol #2

Plane Set $= \left\{ \begin{bmatrix}2\\0\\0\end{bmatrix} + \begin{bmatrix}2\\-4\\0\end{bmatrix} b + \begin{bmatrix}2\\0\\1\end{bmatrix} c \;\middle|\; b,c\in\mathbb{R} \right\}$

$\therefore$ Intersection $= \left\{ \begin{bmatrix}x\\y\\z\end{bmatrix} \in \mathbb{R}^3 \;\middle|\; \begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}2\\0\\0\end{bmatrix} + \begin{bmatrix}2\\-4\\0\end{bmatrix} b + \begin{bmatrix}2\\0\\1\end{bmatrix} c = \begin{bmatrix}0\\2\\0\end{bmatrix} + \begin{bmatrix}1\\3\\1\end{bmatrix} a \text{ where } a,b,c\in\mathbb{R} \right\}$

$$\begin{array}{c} \begin{matrix} a & b & c & \end{matrix} \\ \left[\begin{array}{ccc|c} 1 & -2 & -2 & 2 \\ 3 & 4 & 0 & -2 \\ 1 & 0 & -1 & 0 \end{array}\right] \end{array} \xrightarrow[-r_1+r_3]{-3r_1+r_2} \left[\begin{array}{ccc|c} 1 & -2 & -2 & 2 \\ 0 & 10 & 6 & -8 \\ 0 & 2 & 1 & -2 \end{array}\right] \xrightarrow[\substack{-5R_3+R_2 \\ \frac{1}{2}R_3}]{R_3+R_1} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 1 & \frac{1}{2} & -1 \end{array}\right] \to \begin{array}{c} \begin{matrix} a & b & c & \end{matrix} \\ \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & -2 \end{array}\right] \end{array}$$

$\therefore = \left\{ \begin{bmatrix}0\\2\\0\end{bmatrix} + \begin{bmatrix}1\\3\\1\end{bmatrix} 2 \right\}$

OR $\left\{ \begin{bmatrix}2\\0\\0\end{bmatrix} + \begin{bmatrix}2\\-4\\0\end{bmatrix}(-2) + \begin{bmatrix}2\\0\\1\end{bmatrix}(2) \right\}$

$= \left\{ \begin{bmatrix}2\\8\\2\end{bmatrix} \right\}$ #

Sol #3

Line Equations: $a_i x + b_i y + c_i z = d_i$ where $\begin{bmatrix}a\\b\\c\end{bmatrix} \cdot \begin{bmatrix}1\\3\\1\end{bmatrix} = 0$

$$\begin{array}{c} \begin{matrix} a & b & c & \end{matrix} \\ \left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \end{array}\right] \end{array} \therefore \begin{bmatrix}a\\b\\c\end{bmatrix} \in \left\{ \begin{bmatrix}-3\\1\\0\end{bmatrix} b + \begin{bmatrix}-1\\0\\1\end{bmatrix} c \;\middle|\; b,c\in\mathbb{R} \right\}$$

Let $b=c=1$; แทน $\begin{bmatrix}0\\2\\0\end{bmatrix}$ เพื่อหา $d_i \to d_1=2,\ d_2=0$

$\boxed{\begin{array}{l} -3x+y=d_1 \\ -x+z=d_2 \end{array}} \to \boxed{\begin{array}{l} -3x+y=2 \\ -x+z=0 \end{array}}$

Plane Equation: $\boxed{2x+y-4z=4}$

$$\begin{array}{c} \begin{matrix} x & y & z & \end{matrix} \\ \left[\begin{array}{ccc|c} 2 & 1 & -4 & 4 \\ -3 & 1 & 0 & 2 \\ -1 & 0 & 1 & 0 \end{array}\right] \end{array} \to \left[\begin{array}{ccc|c} 0 & 1 & -2 & 4 \\ 0 & 1 & -3 & 2 \\ 1 & 0 & -1 & 0 \end{array}\right] \to \begin{array}{c} \begin{matrix} x & y & z & \end{matrix} \\ \left[\begin{array}{ccc|c} 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 8 \\ 1 & 0 & 0 & 2 \end{array}\right] \end{array}$$

$\therefore$ Intersection $\begin{bmatrix}x\\y\\z\end{bmatrix}$ must solve both line & plane equations

$\therefore \left\{ \begin{bmatrix}2\\8\\2\end{bmatrix} \right\}$ #

5. (1 point) Find all possible values of x such that $A = \begin{bmatrix}3 & 2 & 4\\1 & 5 & 6\\1 & 2 & x\end{bmatrix}$ is a nonsingular matrix.

$|A| \neq 0$

$[3(5)(x) + 2(6)(1) + 2(4)(1)] - [1(5)(4) + 2(6)(3) + 1(2)(x)] \neq 0$

$[15x + \cancel{12} + 8] - [\cancel{20} + 36 + 2x] \neq 0$

$13x \neq 36$

$x \neq \frac{36}{13}$

$\therefore \boxed{x \in \mathbb{R} - \left\{\frac{36}{13}\right\}}$ #

Name: _____ ID: _____ No: _____

6. (3 points) Let T_1 and T_2 be two linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^n$,
   a. If T_1 and T_2 are not onto transformations, can we conclude that the composition $T_1 \circ T_2$ be an onto transformation? Why?
   b. If T_1 and T_2 are not one-to-one transformations, can we conclude that the composition $T_1 \circ T_2$ be a one-to-one transformation? Why?
   c. Given a transformation T_1 which is not an onto transformation and T_2 which is not a one-to-one transformation, what can we conclude for $T_1 \circ T_2$ and $T_2 \circ T_1$? Why?

☆ For any linear transformation $T: \mathbb{R}^n \to \mathbb{R}^n$, $\boxed{T \text{ is one-to-one iff } T \text{ is onto}}$

Let $A_{n\times n}$ represent T, T is 1-1 $\longleftrightarrow$ RRE(A) has pivot every column

$\updownarrow$ $\updownarrow$ A is square!

T is onto $\longleftrightarrow$ RRE(A) has pivot every row.

Let $n\times n$ matrices A_1, A_2 represent T_1, T_2 respectively.

ⓐ T_1 is not onto, then RRE(A_1) has zeros-row.

* Row Operation is a linear transformation $\therefore$ RRE(A_1) $= MA_1$
where M represents row operations from $A_1 \to$ RRE(A_1)

$T_1 \circ T_2$ is represented by A_1A_2, if we apply row operation M, we have $A_1A_2 \to MA_1A_2$

Since MA_1 has zeros-row, MA_1A_2 must also have zeros-row, and so does RRE(A_1A_2).

$\therefore$ $\boxed{T_1 \circ T_2 \text{ is not onto!}}$ ※

ⓑ From ☆, if T_1 is not 1-1, then T_1 is also not onto.
From ⓐ, if T_1 is not onto, then $T_1 \circ T_2$ is not onto.
From ☆, if $T_1 \circ T_2$ is not onto, then $\boxed{T_1 \circ T_2 \text{ is not one-to-one!}}$ ※

ⓒ From ☆, if T_1 is not onto, then T_1 is not one-to-one, and
if T_2 is not one-to-one, then T_2 is not onto.

From ⓐ, ⓑ, $\boxed{\begin{array}{l} T_1 \circ T_2 \text{ is neither onto nor one-to-one!} \\ T_2 \circ T_1 \text{ is neither onto nor one-to-one.} \end{array}}$ ※

Counter Example is acceptable for ⓐ, ⓑ (at least for me)
↳ Just to show that we CANNOT conclude $T_1 \circ T_2$ as 1-1 or onto.
But ⓒ need a full proof, to show what you want to CONCLUDE.

7. (4 points) Let $A_{4\times4} = \begin{bmatrix} 3 & 9 & 6 & 15 \\ 2 & 4 & 10 & 8 \\ 14 & 7 & 140 & 35 \\ 1 & 2 & 3 & 5 \end{bmatrix}$ and $B_{4\times3} = \begin{bmatrix} 3 & 6 & 9 \\ 2 & 4 & 10 \\ 7 & 21 & 14 \\ 5 & 15 & 10 \end{bmatrix}$, find $A^{-1}B$.

$$[A \mid B] \xrightarrow{\cdots} [A^{-1}A \mid A^{-1}B] = [I \mid A^{-1}B]$$

$$\left[\begin{array}{cccc|ccc} 3 & 9 & 6 & 15 & 3 & 6 & 9 \\ 2 & 4 & 10 & 8 & 2 & 4 & 10 \\ 14 & 7 & 140 & 35 & 7 & 21 & 14 \\ 1 & 2 & 3 & 5 & 5 & 15 & 10 \end{array}\right]$$

$$\xrightarrow[\frac{1}{7}\rho_3]{\frac{1}{3}\rho_1,\ \frac{1}{2}\rho_2} \left[\begin{array}{cccc|ccc} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 1 & 2 & 5 & 4 & 1 & 2 & 5 \\ 2 & 1 & 20 & 5 & 1 & 3 & 2 \\ 1 & 2 & 3 & 5 & 5 & 15 & 10 \end{array}\right]$$

$$\xrightarrow[-\rho_1+\rho_4]{-\rho_1+\rho_2,\ -2\rho_1+\rho_3} \left[\begin{array}{cccc|ccc} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 0 & -1 & 3 & -1 & 0 & 0 & 2 \\ 0 & -5 & 16 & -5 & -1 & -1 & -4 \\ 0 & -1 & 1 & 0 & 4 & 13 & 7 \end{array}\right]$$

$$\xrightarrow[-5\rho_2+\rho_3,\ -\rho_2+\rho_4]{3\rho_2+\rho_1,\ -\rho_2} \left[\begin{array}{cccc|ccc} 1 & 0 & 11 & 2 & 1 & 2 & 9 \\ 0 & 1 & -3 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & -1 & -1 & -14 \\ 0 & 0 & -2 & 1 & 4 & 13 & 5 \end{array}\right]$$

$$\xrightarrow[2\rho_3+\rho_4]{-11\rho_3+\rho_1,\ 3\rho_3+\rho_2} \left[\begin{array}{cccc|ccc} 1 & 0 & 0 & 2 & 12 & 13 & 163 \\ 0 & 1 & 0 & 1 & -3 & -3 & -44 \\ 0 & 0 & 1 & 0 & -1 & -1 & -14 \\ 0 & 0 & 0 & 1 & 2 & 11 & -23 \end{array}\right]$$

$$\xrightarrow{-2\rho_4+\rho_1,\ -\rho_4+\rho_2} \left[\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 8 & -9 & 209 \\ 0 & 1 & 0 & 0 & -5 & -14 & -21 \\ 0 & 0 & 1 & 0 & -1 & -1 & -14 \\ 0 & 0 & 0 & 1 & 2 & 11 & -23 \end{array}\right]$$

$$\therefore A^{-1}B = \begin{bmatrix} 8 & -9 & 209 \\ -5 & -14 & -21 \\ -1 & -1 & -14 \\ 2 & 11 & -23 \end{bmatrix} \#$$